

ZKTrustLLM-Agents Level 4 Technical Report

Steps 77-91: MCP/A2A Reference-Bound Coordination, ZK Proofs, Control-Plane Telemetry, RTP Media Validation, DTLS Protection, and Network Impairment Evaluation

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1. Executive Summary

This report consolidates the Level 4 ZKTrustLLM-Agents workflow from Step 77 to Step 91. The work progresses from blockchain-authenticated A2A references and MCP context retrieval to AUTH_V2.3 reference-bound proof verification, negative-security testing, control-plane telemetry, multi-agent scaling, live RTP media-plane measurement, DTLS-wrapped RTP validation, and network impairment evaluation.

The main research direction is to show how Agentic AI systems can coordinate through compact references to authenticated blockchain state while preserving proof-governed decision semantics and measurable network/media-plane behaviour.

2. Step Timeline and Artifact Map

Step	Main contribution	Primary evidence
77	A2A reference-aware coordination	Reference registry and compact inter-agent reference exchange
79	MCP Level 4 context server	Structured tools for blockchain-authenticated state retrieval
80	MCP/A2A KPI analysis	Agent/control KPI summary
81	AUTH_V2.3 reference-bound proof	Groth16-style proof-bound decision submission
82	Negative-security tests	Tampered input and invalid-context rejection
83	Network KPI telemetry-emulation	Control response, jitter, packet loss, interruption estimates
84	Journal evaluation write-up	Paper-ready Level 4 evaluation sections and tables
85	Benchmark result figures	Agent scaling and utility figures
86	Semi-live control telemetry	Measured MCP/A2A local control-loop timing
87	Multi-agent scaling telemetry	5-25 agent scaling evidence
88	Technical report PDF	Supervisor-ready report up to Step 87
89	Live RTP media-plane telemetry	Packet continuity, RTP jitter, bitrate
90	DTLS-wrapped RTP validation	Secured RTP tunnel/proxy baseline
91	Network impairment evaluation	Plain RTP vs DTLS-RTP under tc/netem stress

3. Security Validation Evidence

Layer	Metric	Result	Interpretation
AUTH_V2.3	Negative cases	7	Tampered proof-bound inputs tested
AUTH_V2.3	Passed negative cases	7	Tampered proof paths rejected
AUTH_V2.3	Tampered rejection rate	1	Verifier/attestor rejects invalid public inputs
MCP/A2A	Invalid-context rejection rate	-	Invalid A2A references rejected

4. Semi-Live MCP/A2A Control-Plane Telemetry

Mode	Runs	Success	Avg ms	P50 ms	P95 ms	Jitter std	Avg bytes
baseline_direct_agent	20	1	0.001	0.001	0.001	0	159.55
l4_raw_context	20	1	72.793	72.974	77.039	3.119	3477.25
l4_ref_mcp_a2a	20	1	34.918	35.556	36.155	1.406	72.55

Step 86 reports L4-ref latency reduction versus L4 raw-context of 52.03% and control-message reduction of 97.91%.

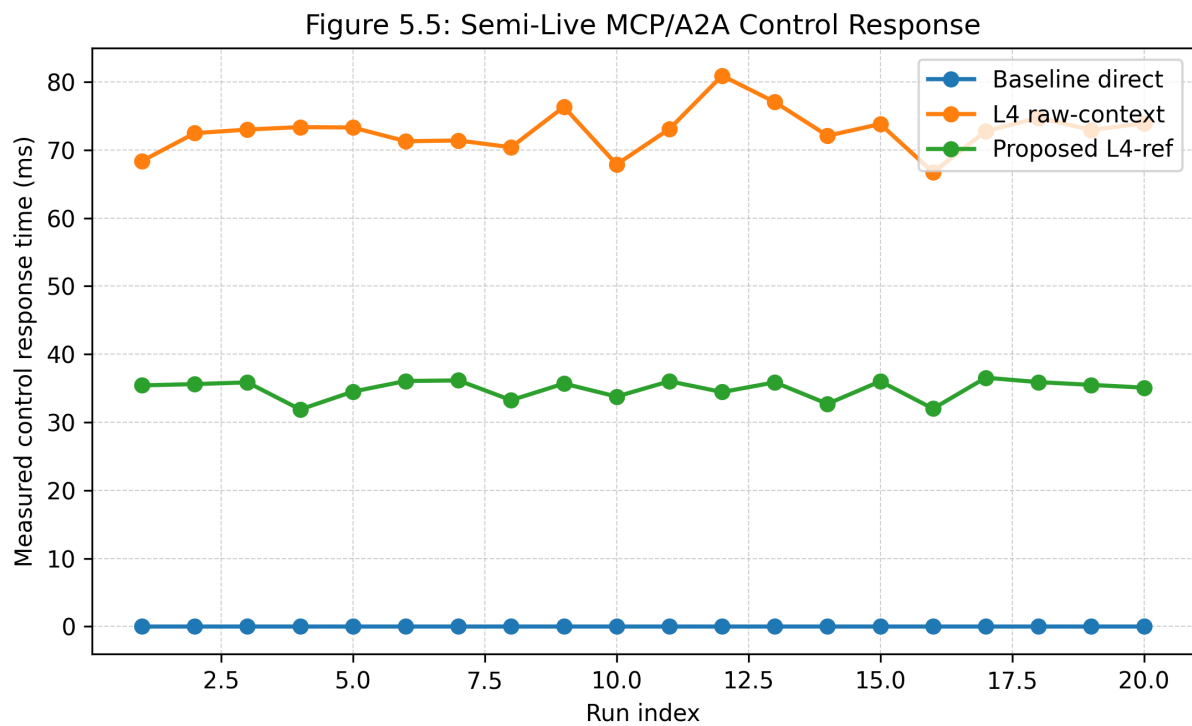


Figure 5.5: Semi-live MCP/A2A control latency.

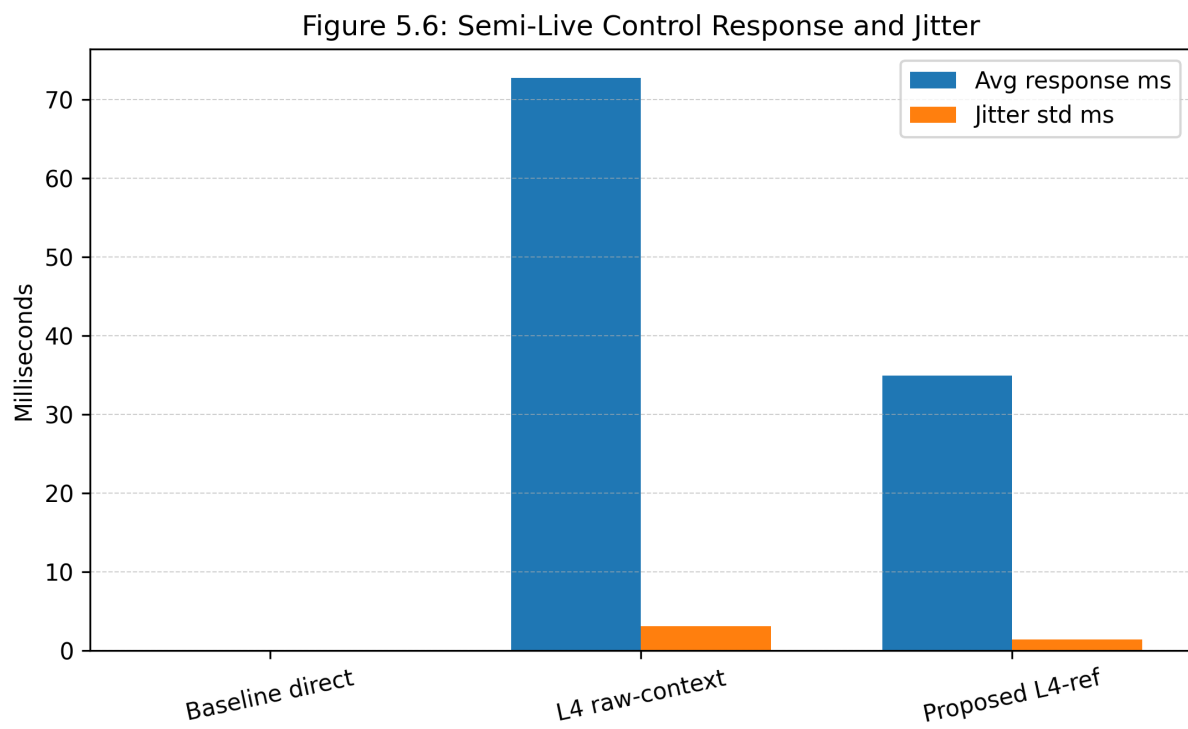


Figure 5.6: Semi-live control summary.

5. Multi-Agent Scaling Telemetry

Mode	Agents	Success	Total ms	Per-agent ms	Bytes	Agents/s
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Step 87 measures how Level 4 MCP/A2A coordination behaves across 5, 10, 15, 20, and 25 agents. The key result is that compact reference exchange significantly reduces control-message size and improves throughput compared with raw-context exchange.

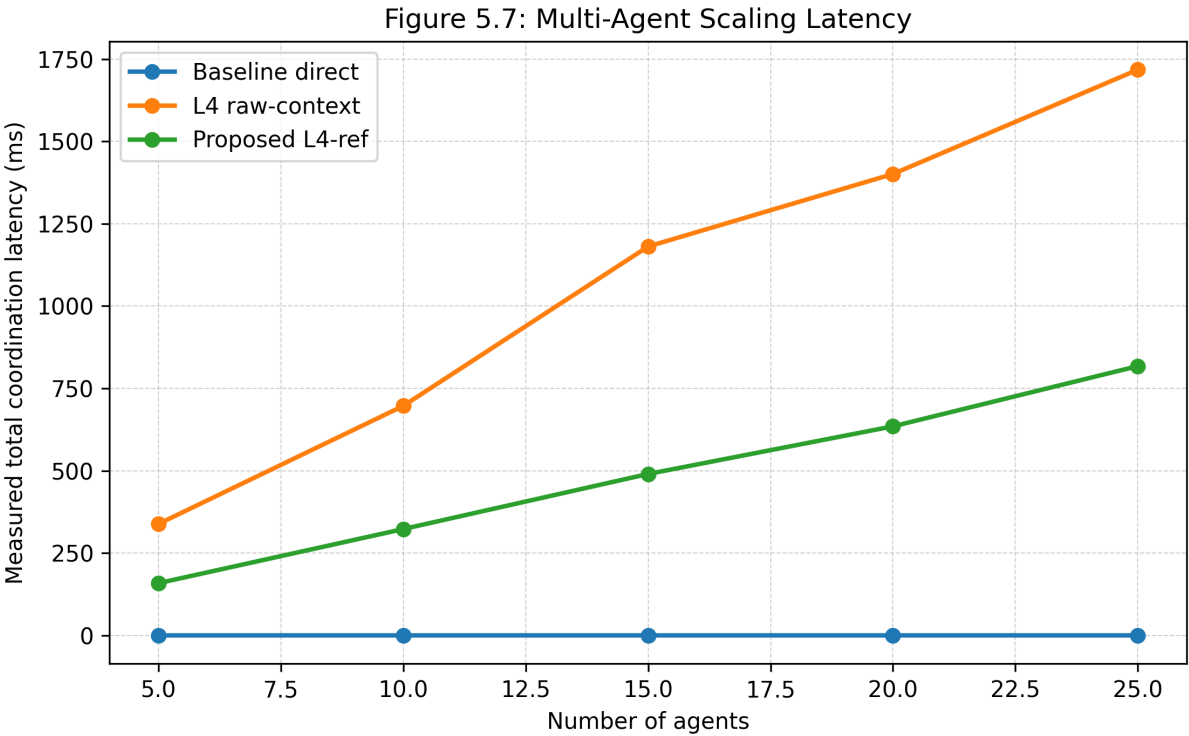


Figure 5.7: Multi-agent scaling latency.

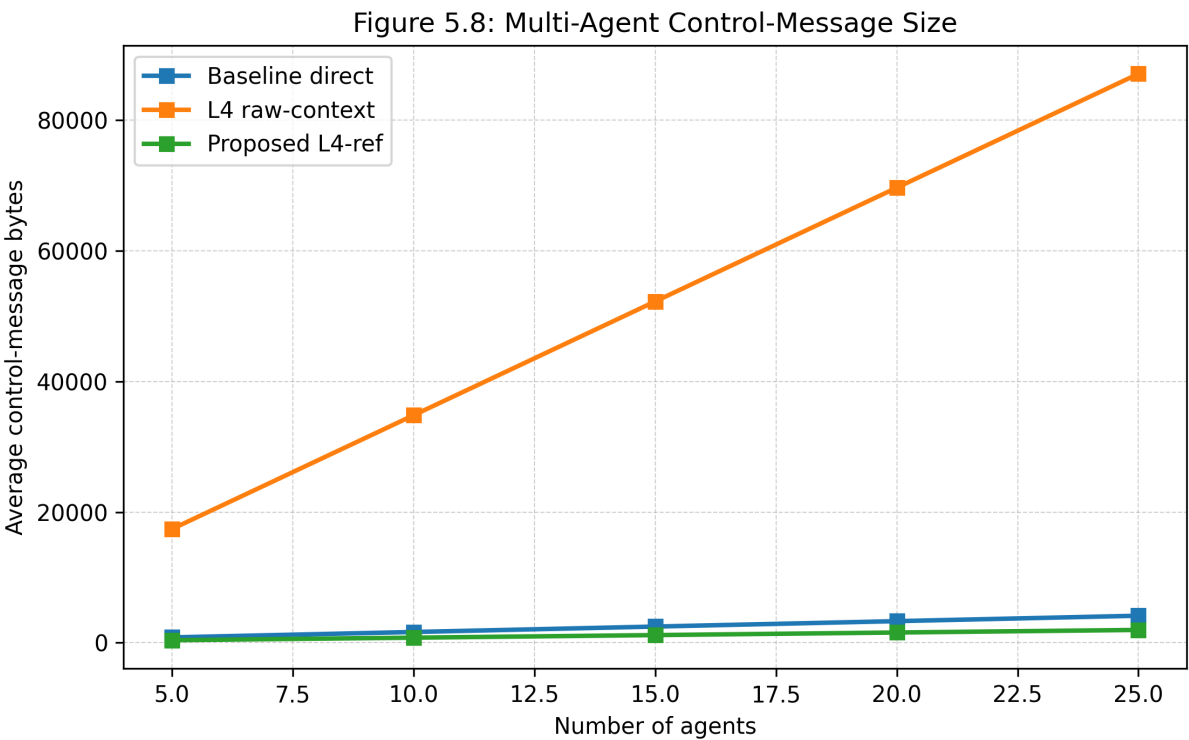


Figure 5.8: Multi-agent control-message size.

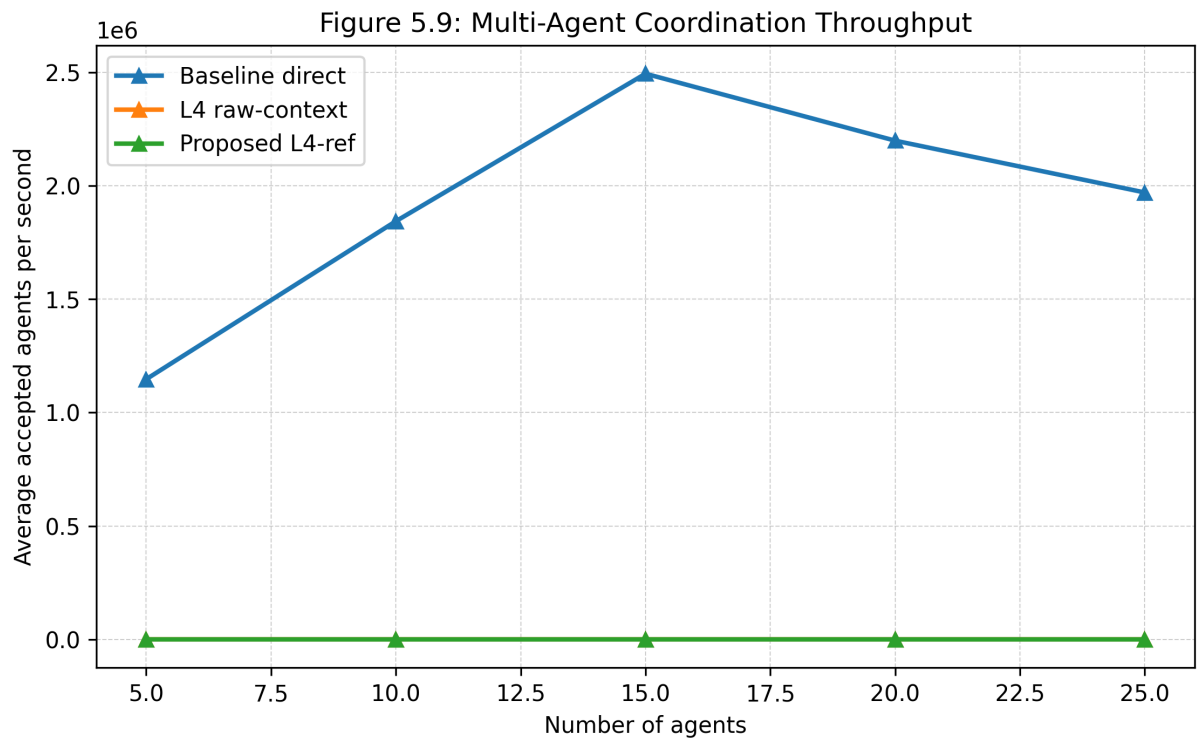


Figure 5.9: Multi-agent throughput.

6. Live RTP Media-Plane Telemetry

KPI	Result	Interpretation
Capture duration	25	Live RTP capture window
Received packets	2328	Captured RTP packets
Expected packets	2328	Expected from RTP sequence continuity
Lost packets	0	Sequence-gap loss
Packet loss %	0	Media-plane packet loss
Average bitrate kbps	833.605	RTP throughput
Average jitter ms	0.229169	RTP timing variation
P50 jitter ms	0.069357	Median timing variation
Max jitter ms	4.16395	Largest observed deviation

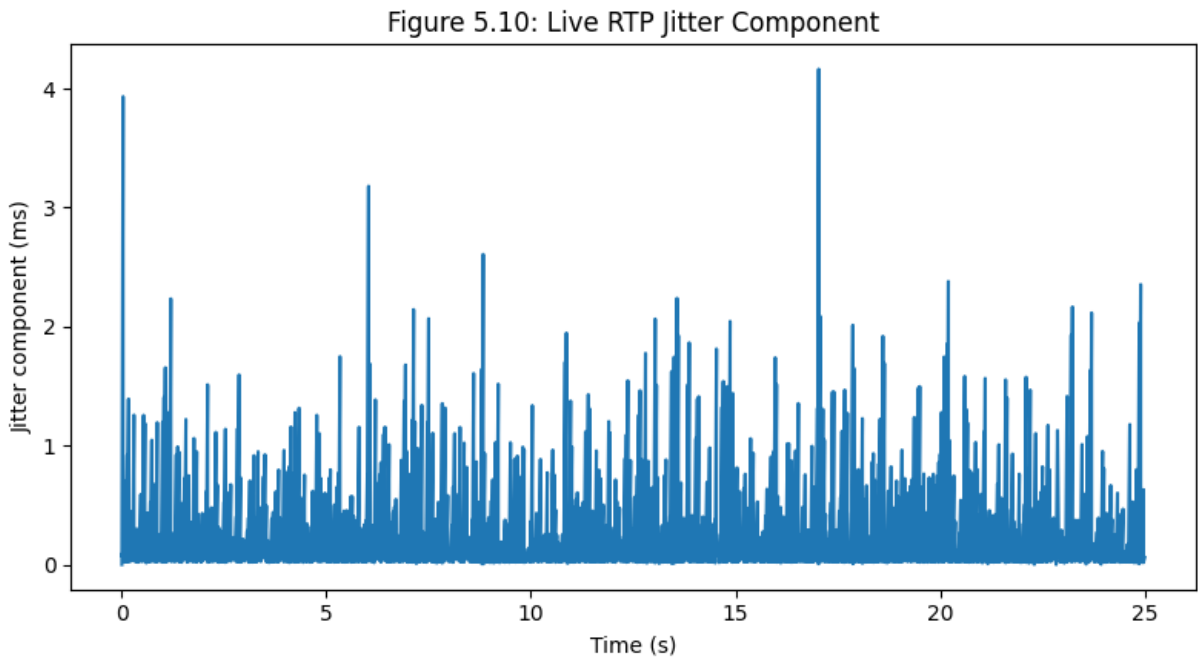


Figure 5.10: Live RTP jitter.

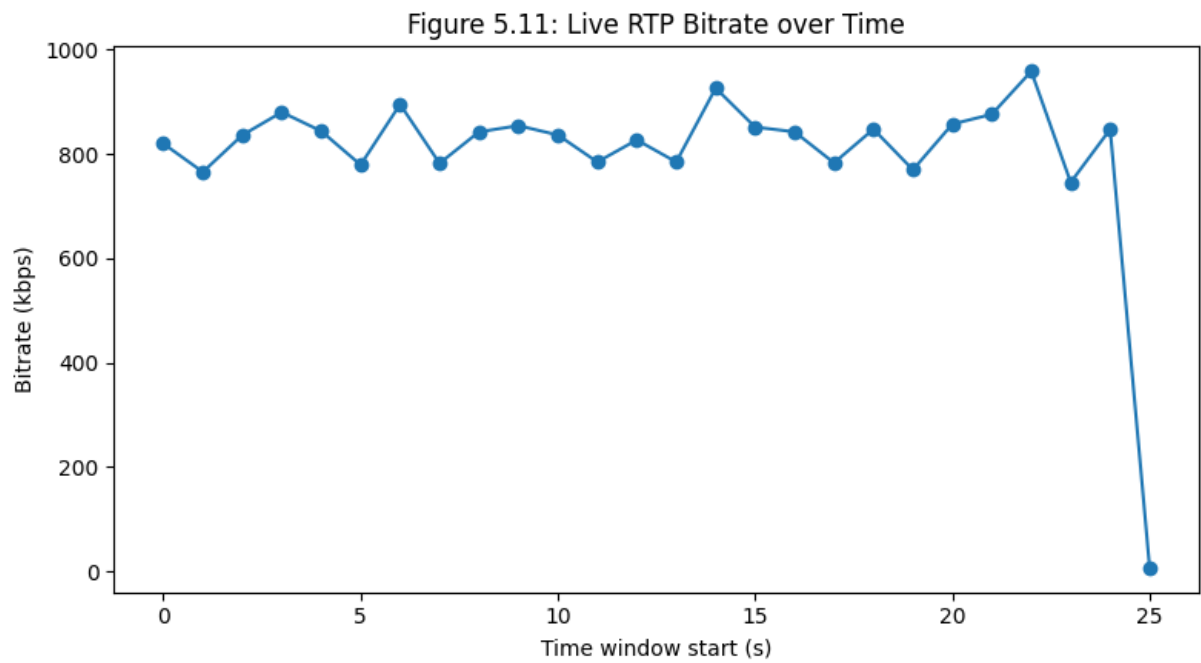


Figure 5.11: Live RTP bitrate.

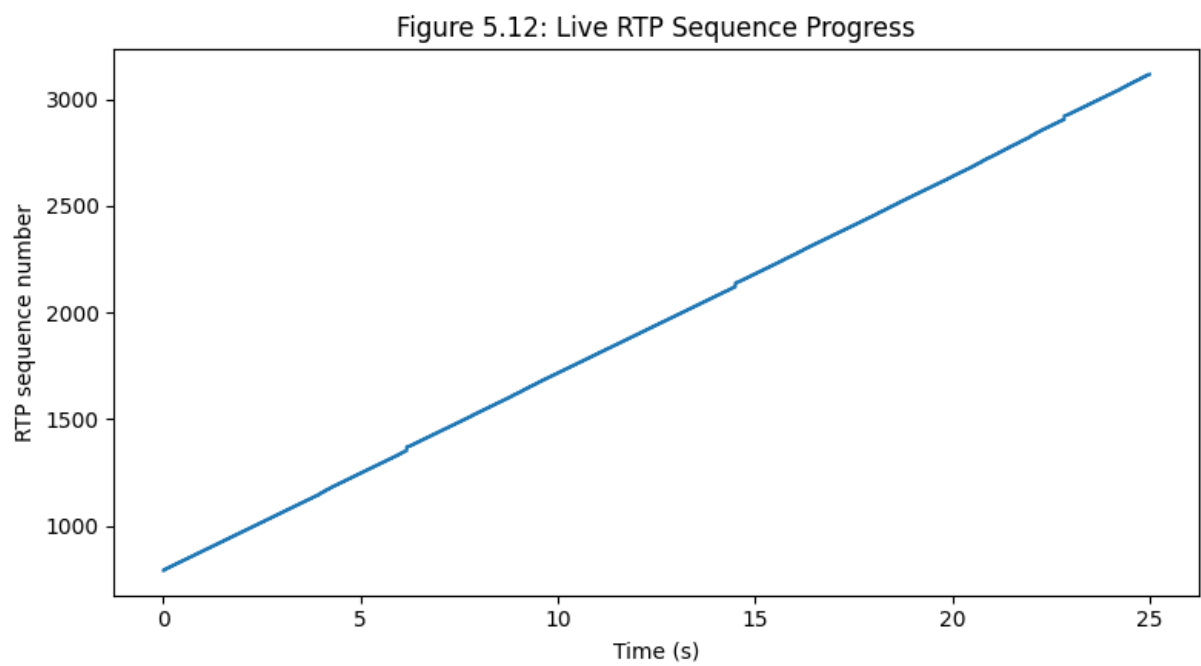


Figure 5.12: RTP sequence progression.

7. DTLS-Wrapped RTP Media-Plane Validation

KPI	Result	Interpretation
Recovered RTP packets	769	Packets recovered after DTLS tunnel
Expected packets	769	Expected sequence continuity
Lost packets	0	Sequence-gap loss
Packet loss %	0	DTLS-wrapped RTP loss
Average bitrate kbps	102.17	Recovered media throughput
Average jitter ms	0.95001	Arrival-gap timing variation
P50 jitter ms	0.1185	Median timing variation
Max jitter ms	33.3165	Largest observed deviation

Step 90 validates a DTLS tunnel/proxy baseline for RTP. It is not WebRTC DTLS-SRTP, but it demonstrates that RTP packets can be protected, recovered, and measured after DTLS transport.

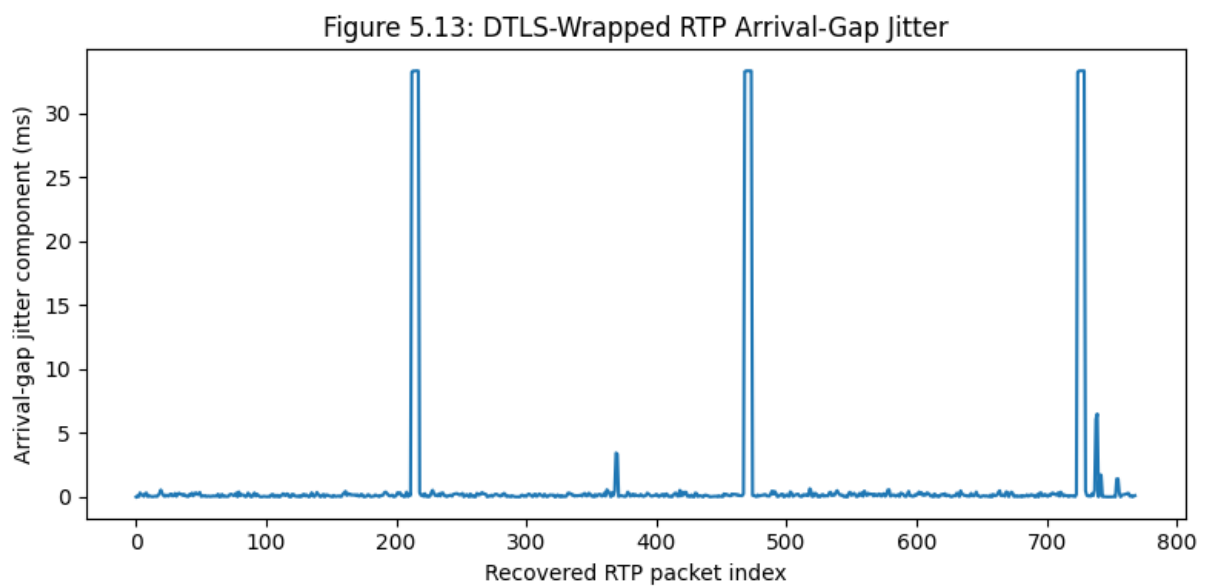


Figure 5.13: DTLS-wrapped RTP jitter.

Figure 5.14: Plain RTP vs DTLS-wrapped RTP packet loss

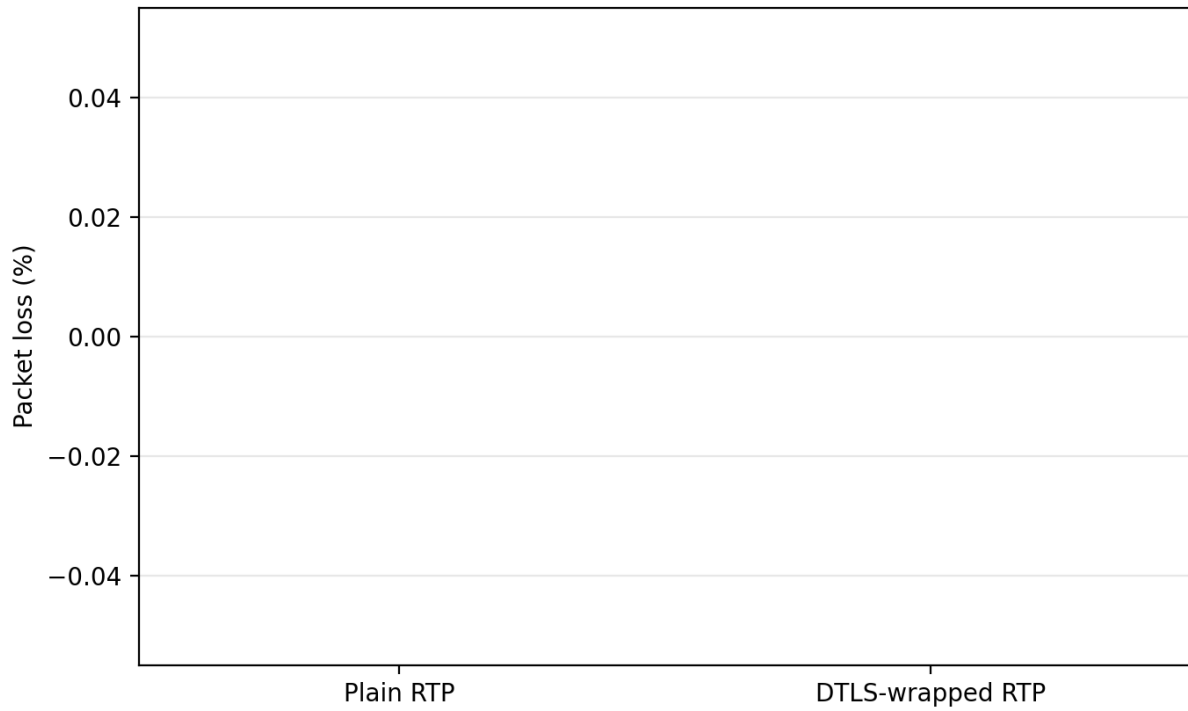


Figure 5.14: Plain RTP vs DTLS packet loss.

Figure 5.15: Plain RTP vs DTLS-wrapped RTP bitrate

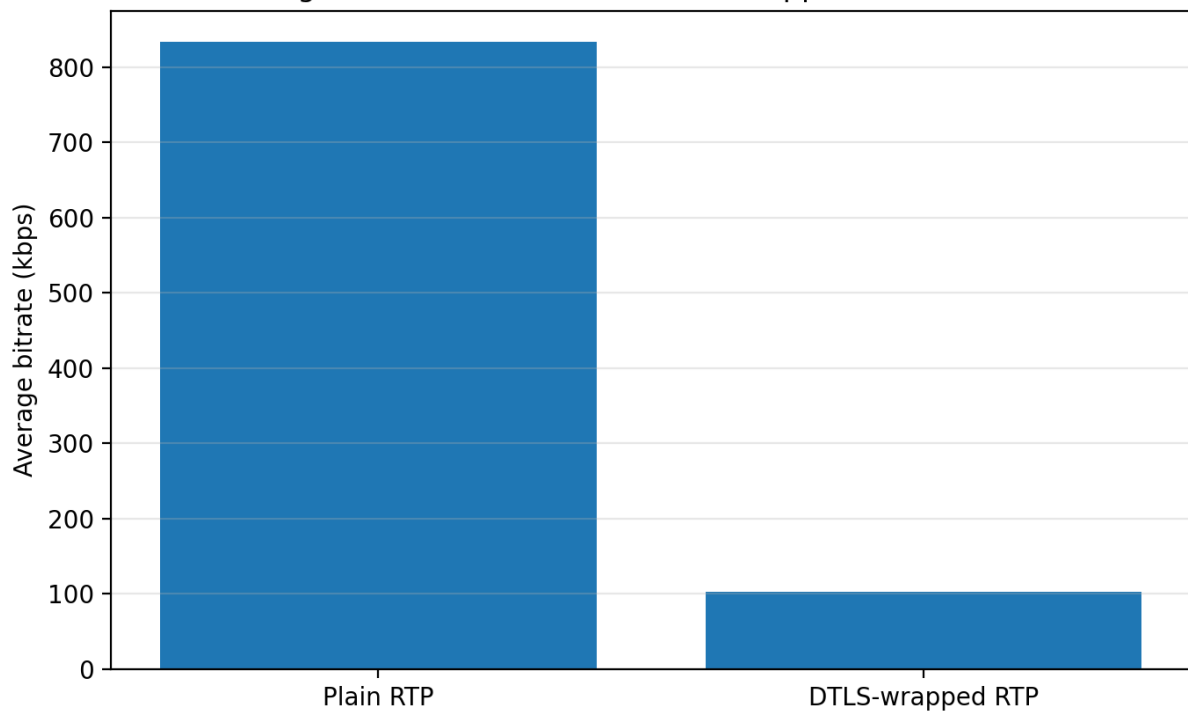


Figure 5.15: Plain RTP vs DTLS bitrate.

8. Network Impairment Evaluation: Plain RTP vs DTLS-RTP

Step 91 compares plain RTP and DTLS-wrapped RTP under controlled loopback impairment using Linux tc netem. Profiles include clean baseline, fixed delay, delay with jitter, and delay with jitter plus packet loss. Jitter is reported as packet arrival-gap jitter to avoid misleading raw RTP timestamp artefacts.

Profile	Mode	Packets	Loss %	Bitrate kbps	Avg jitter	P50	P95
clean_baseline	plain_rtp	2328	0	833.605	10.7091	0.045	34.019
clean_baseline	dtls_rtp	769	0	102.17	0.926337	0.1165	0.4245
delay_20ms	plain_rtp	2328	0	833.605	10.7031	0.057	33.846
delay_20ms	dtls_rtp	769	0	102.419	0.98316	0.1665	0.5325
delay_20ms_jitter_5ms	plain_rtp	2328	28.6983	833.709	8.32631	4.855	24.453
delay_20ms_jitter_5ms	dtls_rtp	769	2.41117	102.419	10.2768	8.8005	24.8545
delay_30ms_jitter_10ms_loss_1pct	plain_rtp	2300	28.5492	823.692	6.97549	5.834	19.719
delay_30ms_jitter_10ms_loss_1pct	dtls_rtp	737	13.7002	98.3696	17.4648	15.8885	37.8075

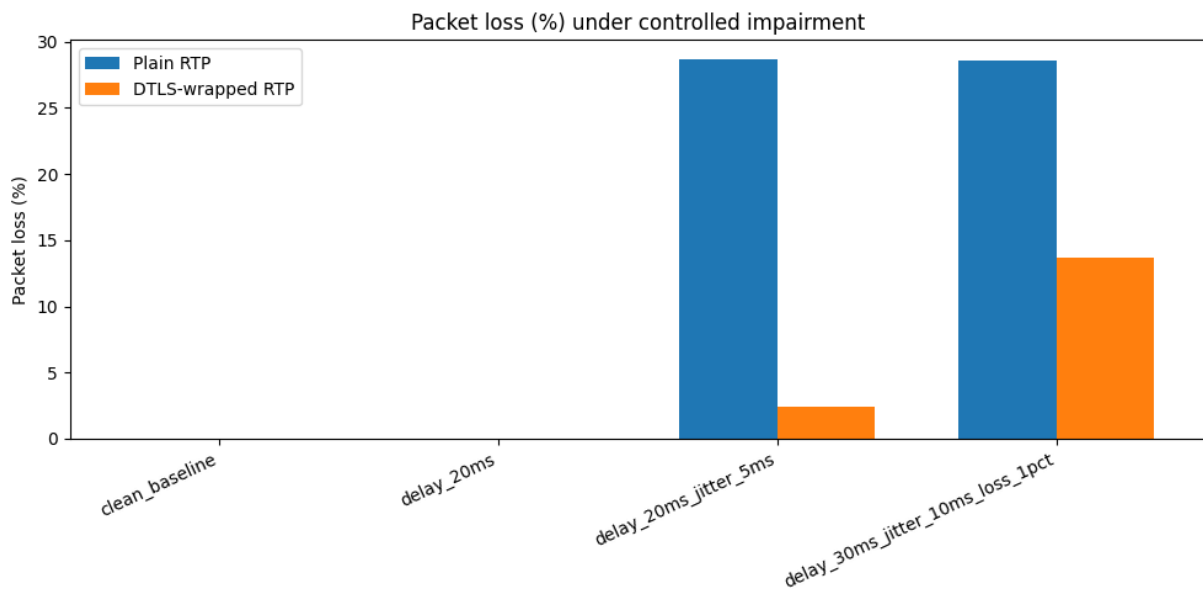


Figure 5.16: Packet loss under impairment.

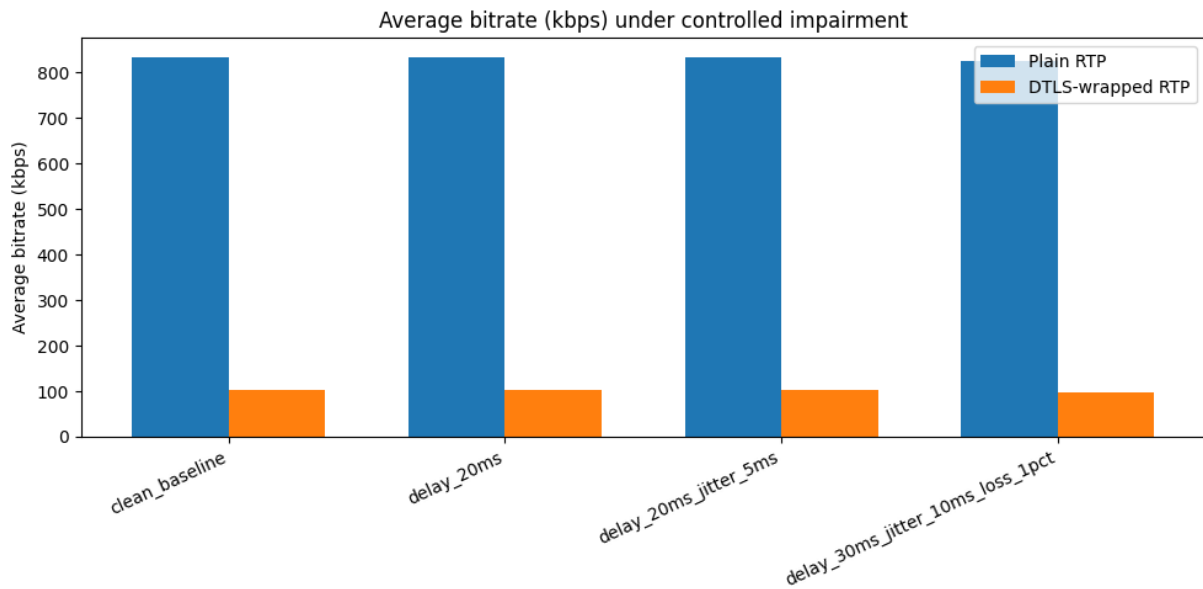


Figure 5.17: Bitrate under impairment.

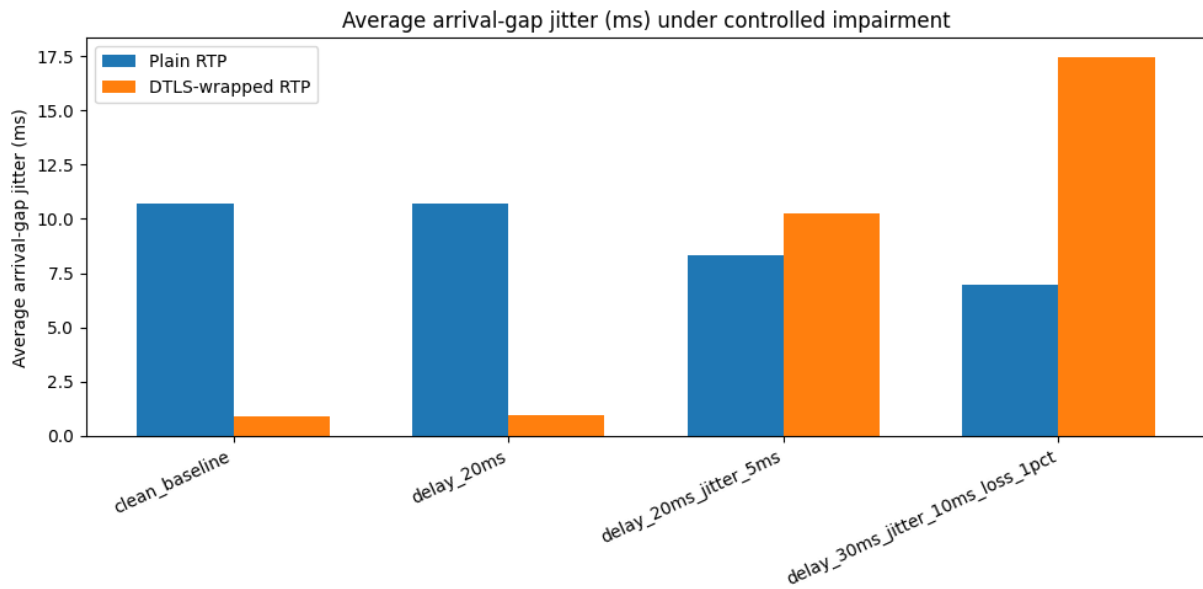


Figure 5.18: Arrival-gap jitter under impairment.

9. Journal-Ready Interpretation

The Level 4 implementation now contains a complete evaluation ladder: proof-bound decision generation, negative-security rejection, MCP/A2A reference verification, semi-live control-plane telemetry, multi-agent scaling, live RTP media-plane measurement, DTLS-wrapped RTP validation, and impairment testing. This supports a journal-level argument that blockchain-authenticated shared state can be used by Agentic AI systems through compact A2A references and MCP tools, while media-plane KPIs can be measured under secure and impaired delivery conditions.

The current limitations are also clear: the experiments run on localhost, the DTLS design is a controlled tunnel/proxy rather than WebRTC DTLS-SRTP, and future validation should move to two-machine, network namespace, Mininet, ns-3, or 5GIC testbed scenarios.

10. Recommended Next Steps

Priority	Next step	Reason
1	Two-machine RTP/DTLS validation	Move beyond localhost loopback
2	Network namespace or Mininet replication	Cleaner impairment isolation and reproducibility
3	Integrate LKH policy triggers with media experiments	Connect proof-governed control decisions to media-plane response
4	Add repeated trials and confidence intervals	Strengthen statistical validity
5	Prepare journal Results section	Use Step 86-91 tables and figures directly